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PROJECT AND TEAM INFORMATION

Project Title

College Event Registration Platform

Student / Team Information

Team Name: *Team ID:*	**BRAWL CODEs** **DSCPP-III-2026-T188**
Team member 1 (Team Lead)	*Nikunj Singhal – 2510370169* *2510370169@geu.ac.in* *University Roll No.: 2028783* *Phone No.: 9389549348* *Section: M1-3* 
Team member 2	*Bhargav Istwal – 2510371598* *2510371598@geu.ac.in* *University Roll No.: 2028688* *Phone No.: 8171097056* *Section: ML-2* 
Team member 3	*Harshit Kashyap – 2510011536* *2510011536@geu.ac.in* *University Roll No.: 2028724* *Phone No.: 9882847346* *Section: ML-4* 

Project Mentor: Dr. Prabhdeep Singh **Interactions completed:** 02

PHASE-III PROJECT PROGRESS DESCRIPTION

Project Abstract

The College Event Registration Platform is a console-based application developed to make college event management and student registration easier and more organized. The main goal was to provide students with a simple way to explore events
register for them and manage their registrations while allowing administrators to monitor events and student records. Our final version includes separate student and admin login systems event searching, registration and cancellation, duplicate registration prevention, waiting-list management and event sorting It uses C++ classes and objects, a singly linked list for managing events, queues for waiting lists and a separate stack module to demonstrate basic operations Student details, registration records and waiting-list information are stored in local text files using file handling The application was tested with different scenarios, including login validation, duplicate registration, event searching, sorting and data saving. the project demonstrates how C++ programming. The concept of data structures and object oreinted programming is used for developing this college event management system.

Updated Project Approach and Architecture

Our C++ application is divided into student and administrator modules, with separate menus which handles their operations. The Student module allows users to log in to system and view events, they can search by name or category also register for events, cancel registrations and check their waiting list details. The admin module allows administrators to view events student records, registrations and waiting lists, save all data and sort events by name or available seats. We use a singly linked list to manage records of events, a queue is used to maintain waiting lists and a separate stack module to demonstrate push, pop and other operations. searching sorting, traversal and dynamic memory allocation are also used for process record.
The application uses C++ classes such as student and event to organize data and functions. private and public access specifiers help us to control access to class members, while constructors initializes object. Encapsulation for data hiding and abstraction are used to build the program. File handling is implemented through text files such as students.txt registrations.txt
waiting lists.txt to store and retrieve application records. Event information is initialized with the program. We developed here the application using C++ and tested its major operations through different input based scenarios, including login validation, duplicate registration prevention, event search, sorting and data saving. The system uses local text files and does not currently depend on an external web interface.

Tasks Completed

Task Completed	Team Member
Responsible for project planning, team coordination and organizing the overall development process.	Nikunj Singhal (ML-3)
Responsible for C++ development, implementation of data structures and algorithm and project documentation.	Bhargav Istwal (ML-2)
Responsible for testing the application, checking its functionality and verifying systems through different senarios.	HarshitKashyap (ML-4)

Challenges/Roadblocks

One of the main challenges was managing event and student registration records while keeping the available seat count accurate preventing duplicate registrations was also important so the system checks whether a student is already registered before accepting a new request managing waiting lists required maintaining the correct order of students using a FIFO queue. another challenge was handling invalid inputs incorrect login credentials and unsuccessful searches without interrupting the application we also had to manage dynamically allocated memory while working with linked lists and queues. File handling required proper reading and writing of student, registration and waiting list records so that information could be retained between sessions. these challenges were addressed through input validation, suitable data structures, separate student and administrator modules, and file-handling operations. Further testing is still required for waiting-list promotion, data persistence after restarting the application and different edge cases.

Project Outcome/Deliverables

The final outcome is a working C++ console-based College Event Registration Platform with separate student and administrator modules students can view events, search by name or category register for events, cancel registrations, check their registrations and view waiting lists. administrators can view student details, event registrations, available seats and waiting list records, as well as sort events and save application data. The program is built around C++ and uses different concepts to handle its operations. We used a linked list to maintain event detail and a queue which keep track of student waiting for a seat. A separate module covers stack operation. File handling is used to save student information, registration details and waiting list records. Along with the source code, our final submission contains the required text files, project report, testing details and presentation. We have also uploaded the project to GitHub. These materials show the work completed during the development of the College Event Registration Platform.The work on this project was divided into three phases. At the beginning, we chose the problem of managing college events and discussed what features would be needed in the application. We then in second phase planned its basic structure and core dsa and c++ concepts. At phase 3 we have developed the console based system.

Progress Overview

Our College Event Registration Platform was developed in three phases. In Phase I we selected the problem, discussed the requirements and planned the basic structure of the application. During Phase II we worked on the C++ program and developed its main features using OOP concepts and data structures. In Phase III we completed the student and admin modules, worked on registration management, tested the application and prepared the final documentation The program now allows students to search for events, register or cancel their registrations and view waiting lists. It also prevents duplicate registrations and provides options for sorting events. We tested the main features with different inputs to check whether they were working correctly. Some parts, such as automatic waiting-list promotion, data loading after restarting and certain data-structure operations, still need more testing. Features like a graphical interface, online registration and database connectivity could be added in the future.

Codebase Information

Repository: https://github.com/bhargavistwal/College-Event-Registration-Platform

Branch: main.
The repository contains the complete project files, which includes C++ source code with supporting modules and docs. It is organized to keep the main application and different functionalities easy to access. The source code covers all things such as event management, student registration, waiting-list operations, searching and sorting. It also includes the text files used which store student details, registration records etc. The Readme file provides details about the project and how to run it. GitHub is used as a platform to deploy all the files and repositries which we have worked on.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Student and Admin Login	Pass	Both login systems are tested succesfuly.
Duplicate Registration	Pass	Repeated registration for same event is prevented.
Event Searching	Pass	Search event by name / category and unsuccesful search is tested.
Event Sorting	Pass	Event sorted by name and by available seat.
File Handling	Pass	Application data saved succesfuly in text files.
Input Validation	Pass	Invalid credentials and menu input is handled correctly.

Deliverables Progress

Completed: We have finished our C++ console-based College Event Registration Platform with separate options for students and administrators. The project has management of events and the registration and cancellation, also check duplicate registration , waiting list check and file handling. We also worked with linked lists, queues, stacks, searching, sorting and OOP concepts. The main program, testing work, all the reports and presentation and GitHub files have been prepared till now.

Verified: We checked student and admin login, tried registering for an event more than once, event searching and sorting tested, entered incorrect inputs and checked whether application data was saved or not. The main features was demonstrated by the help of console based system, and all planned demonstration modules have been covered.